CAN YOUR AI TUTOR SENSE YOUR BRAIN DRAIN?

Applying Cross-Task EEG Workload Classification in an AI-Supported Learning Environment

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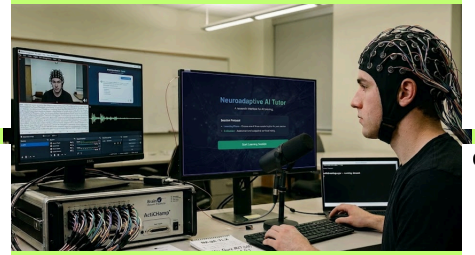
INTRODUCTION

- **The Problem:** Current AI voice tutors are "open-loop" systems; they cannot "see" if a learner is overwhelmed or bored because they lack access to internal cognitive states.
- **The Role of Workload:** Cognitive workload is critical for learning. Excessive demand impairs comprehension, while too little reduces engagement.
- **Neural Markers:** EEG offers a non-invasive solution. High mental effort is reflected by established markers: increased frontal theta and decreased parietal alpha activity [1].
- **The Research Gap:** While EEG classifiers can generalize across structured tasks [2], their use in conversational, LLM-based tutoring is still limited [3, 4].
- **Our Objective:** We developed an LLM tutoring paradigm and conducted experiments to evaluate if a classifier trained on mental arithmetic can accurately detect workload during AI tutoring , specifically analyzing the impact of content complexity and speech pacing.

METHODS

Phase I (Calibration), a mental arithmetic task (serial subtraction) to induce high cognitive workload, contrasted with a low-load condition (eyes-open rest). This data was used to train subject-specific workload classifiers.

Phase II (AI Tutoring), a conversational learning task with an AI tutor, covering 1 minute explanations of three topics (Photosynthesis, Business, Climate). Content Complexity (Simple vs. Complex) and Speech Pacing (Normal vs. Fast) was varied.



Data Acquisition / Tools:

- 64-channel ActiCHamp system at 500 Hz.
- Lab Streaming Layer (LSL) and photodiode triggers for precise event timing.
- Conversational Gemini 2.5 Flash via Google AI Studio.
- Multimodal recording (audio, screen, responses) via OBS Studio.

Cognitive workload assessment:

- EEG-based classifier output (continuous workload estimates)
- Subjective workload ratings (NASA-TLX)
- Comprehension performance (multiple-choice questions following each condition)

DATA PROCESSING

1.EEG PREPROCESSING

- Resample to 250 , Rereference to common average reference.
- Highpass Filter
- Artifact removal using Independent Component Analysis (ICA)
- Components classified with ICLabel; non-brain activity removed (≥90% threshold)

2.BCI MODEL CALIBRATION

- Subject-specific workload classifier [1]
- Feature extraction using Filter Bank Common Spatial Patterns (FBCSP)
- Classification via Linear Discriminant Analysis (LDA)
- Trained on 1-second EEG windows non-overlapping

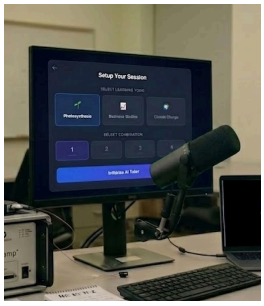
3.CLASSIFICATION ON PARADIGMS:

- Calibrated model applied offline to AI tutoring EEG data.
- Continuous data segmented into 1-second windows.
- Output: Mean LDA scores (−1 = low workload, +1 = high workload)

4.EVALUATION:

- Workload estimates compared across content complexity and speech pacing conditions.
- Subjective workload (NASA-TLX) analysed using condition-wise mean scores.
- Comprehension performance assessed via accuracy on multiple-choice questions.

EXPERIMENTAL DESIGN



1. Topic Selection: Participants select one of three topics.
2.Interactive Session: A 1-minute conversation starts and interrupt to ask questions.
3. Comprehension Check: Understanding is measured via a multiple-choice quiz following the lesson.
4. Workload Assessment: Participants rate their mental effort using a NASA-TLX form.
5. Data Recording: Includes Comprehension score, Average workload, and Session logs, is exported as a JSON file for analysis.

RESULTS

Cross-validated accuracy for Phase I : 68.50%

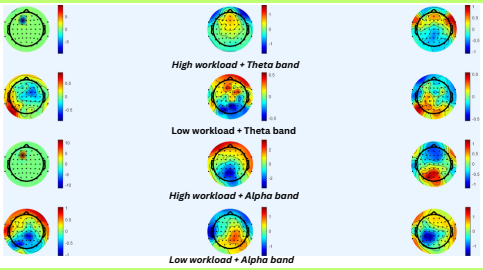
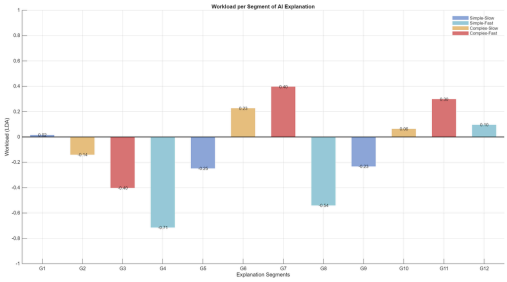
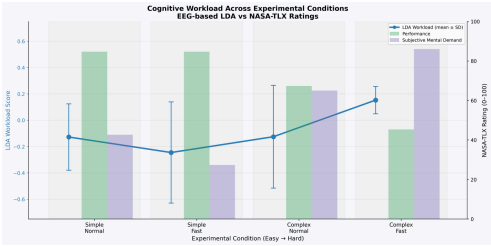


Fig 1: *Spatial patterns of the task-independent predictive model.*

SUBJECTIVE WORKLOAD PLOTS

- *Workload varies clearly across different task conditions*
- *Fast and complex tasks tend to show higher workload levels*
- *Simpler tasks generally show lower or negative workload scores*
- *Workload tends to be higher in more complex tasks*
- *Faster conditions seem to increase mental load*
- *EEG results show similar trends to NASA-TLX ratings*





DISCUSSION

- One of the most interesting findings was that just "going fast" didn't actually tax the brain that much. The EEG workload stayed low during the Simple Fast tasks. It was only when the task got Complex that the brain really had to work.
- When the EEG showed a high workload score, the participant's performance dropped and they reported feeling much more stressed. This suggests the LDA score is a reliable way to track a person's real-time mental breaking point.
- The Complex Normal results show that people can handle a tough task if they aren't being rushed. The participant felt challenged but wasn't overwhelmed until we added the "Fast" element.

CONCLUSION

- **Workload Transfer:** Pilot data (n=1) suggests feasibility of cross-task EEG workload classification in AI tutoring.
- **Key Finding:** Workload was primarily influenced by content complexity; pacing effects were minimal.
- **Implication:** Content adaptation may be more relevant than pacing in neuroadaptive tutoring.
- **Limitation & Future Work:** Single-participant study; future work should use larger samples and more multimodal calibration tasks (beyond arithmetic) for improved generalisation.

REFERENCES

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